

Development of a fully automated RULA assessment system based on computer vision

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ABSTRACT

The purpose of this study was to develop an automated, RULA-based posture assessment system using a deep learning algorithm to estimate RULA scores, including scores for wrist posture, based on images of workplace postures. The proposed posture estimation system reported a mean absolute error (MAE) of 2.86 on the validation dataset obtained by randomly splitting 20% of the original training dataset before data augmentation. The results of the proposed system were compared with those of two experts' manual evaluation by computing the intraclass correlation coefficient (ICC), which yielded index values greater than 0.75, thereby confirming good agreement between manual raters and the proposed system. This system will reduce the time required for postural evaluation while producing highly reliable RULA scores that are consistent with those generated by manual approach. Thus, we expect that this study will aid ergonomic experts in conducting RULA-based surveys of occupational postures in workplace conditions.

1. Introduction

Work-related musculoskeletal disorders (WMSDs) are injuries to the limbs of workers induced or aggravated by working conditions (Schneider et al., 2010). The working conditions that may lead to WMSDs include routine lifting of heavy objects, daily exposure to whole body vibration, routine overhead work, work with the neck in chronic flexion position, or performing repetitive forceful tasks (Bernard, 1997). According to researchers from the National Reference Center for Rare Autoimmune Diseases at University Hospitals of Strasbourg, WMSDs are ranked second worldwide in shortening people's working years, preceded only by mental illness and substance abuse (Sebbag et al., 2019). WMSDs also account for 4.1 million early deaths in 2015, an increase of 46% since 2000 (Sebbag et al., 2019). WMSDs have contributed to almost 400,000 injuries, costing industries over \$20 billion per year (Middlesworth, 2020). Previous studies have identified strong associations between WMSDs and awkward postures in the workplace (Anderson et al., 1997; Simoneau et al., 1996; Van Wely, 1970). Various postural evaluation tools such as the Ovako Working Posture Analysis System (OWAS) (Karhu et al., 1977), the Novel Ergonomic Postural Assessment Method (NERPA) (Sanchez-Lite et al., 2013), the Rapid Upper Limb Assessment (RULA) (McAtamney and Corlett, 1993) and the Rapid Entire Body Assessment (REBA) (Hignett and McAtamney, 2000)

have been developed to evaluate postural risks associated with workplace tasks. These tools are used for initial screenings of postures that can lead to WMSDs and that may require further analysis with more comprehensive tools. RULA is an observational survey method developed to assess postures of the neck, trunk, upper arm, lower arm, and wrists (McAtamney and Corlett, 1993). It needs no special equipment, which makes it easy for investigators to use. The RULA survey tool has proven useful in postural assessments of such occupational fields and setting as supermarkets (Ryan, 1989), agriculture (Tuure, 1992), ship maintenance (Van Wendel de Joode et al., 1997), soft drink distribution (Wright and Haslam, 1999), metalworking (González et al., 2003), transport driving (Massaccesi et al., 2003), carpet mending (Chooibineh et al., 2004), etc. Furthermore, RULA allows for quick assessment of the upper body, making it popular and reliable in industry (Kee, 2020; Kong et al., 2018). Present methods of evaluating postural risk are based on observational techniques that requires an ergonomic analyst observe the work in real-time or from recorded video to manually segment the relevant body parts and evaluate risk associated with the posture (Andrews et al., 2012). Due to human error, however, these techniques produce results with low consistency and repeatability, both of which can be reduced or eliminated by using advanced technologies (Peppoloni et al., 2015; Plantard et al., 2017; Sasikumar and Binoosh, 2020; Clark et al., 2012).

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Human Pose Estimation has significantly progressed with the advancement of Convolution Neural Networks (CNN) and popular keypoints datasets such as Microsoft Common Objects In Context (COCO) (Lin et al., 2014), MPII Human Pose Dataset (Andriluka et al., 2015) and Human 3.6M (Ionescu et al., 2014). The use of objective measurements in postural assessment has been very popular because it eliminates the need for experts to manually segment body parts and evaluate movements and because technological advancements in the field of computing have made it easier than ever to access these tools (Lowe et al., 2019). Earlier attempts to achieve this were based on wearable devices (Peppoloni et al., 2015) and Kinect based systems (Plantard et al., 2017) for online assessment of WMSD risks.

In order to track upper limb movements for calculating work cycles, Peppoloni et al. (2015) used a wireless, wearable device system that relies on EMG signals to determine muscle effort intensity and on inertial measurement units (IMUs) to reconstruct the posture of the human upper limb. This method requires IMU and EMG sensors to be mounted on a subject's body, but sensor application in real work conditions can be difficult due to signal interference, and trained professionals are required to conduct the study and run calibration procedures, which can also be a challenging process, as threshold parameters of the system vary with a subject's motion during calibration (Peppoloni et al., 2015).

Another method uses marker-less motion capture systems like Microsoft Kinect, which are easy-to-use motion capture devices that can provide real-time anatomical landmark position data in three dimensions (Clark et al., 2012). A significant drawback to this method is its occlusion of a body's joints, which can lead to insufficient information to accurately predict posture and hence, unrealistic results (Clark et al., 2012). Another disadvantage of this method is that it requires wrist, wrist twist and neck twist RULA scores to be disregarded because the industrial environment produces too much noise and can lead to error in RULA computation.

A third approach to evaluating postural risk with respect to RULA is to use a computer vision algorithm that predicts the RULA grand score from images. Sasikumar and Binoosh (2020) compared the performance of popular supervised machine learning classifiers such as the Random Forest algorithm, the Naïve Bayes Classifier, the Decision Tree algorithm, the k-Nearest Neighbors algorithm, and Neural Networks and Support Vector Machines when predicting the risk of WMSDs in computer professionals considering postural, physiological, and work-related factors. The attributes for evaluating postural risks have been selected based on data collected from the Nordic musculoskeletal questionnaire survey (Crawford, 2007). Because this study was conducted among computer professionals, it is task-specific and therefore applicable only to IT industries. Li et al. (2020) conducted a study using a Convolution Neural Network (CNN) to predict kinematic joint information from images and another dense network to classify the output as RULA grand scores in real-time. This method uses the MS COCO dataset as training data for a pose detector model to predict key joint locations for workers. One major drawback of this technique, however, is that it assumes a uniform posture score for the wrist. The pose detector model is also sensitive to visual noise in images such as poor lighting conditions and dust, which makes the application difficult in on-site environments.

In general, although direct methods such as motion capture systems or IMU sensors are more accurate than observation-based approaches, they also have the disadvantages of being expensive, requiring training, and being difficult to implement for practitioners in the field. In addition, all the previous studies related to automating the task of posture assessment either require additional mounted equipment or consider the wrist score uniform for group-A postural evaluation and cannot justifiably be applied broadly to general occupational and industrial tasks (Peppoloni et al., 2015; Plantard et al., 2017; Clark et al., 2012; Li et al., 2020). Thus, the proposed study aims to automate the observation-based process of employing RULA in a workplace by developing a deep learning algorithm that can predict full body posture, including wrist posture, from images and evaluate the postural risks associated with

occupational tasks. The present study applied this method to common occupational postures and compares the resulting RULA scores with those calculated by two ergonomics experts. This method reduces the time required for RULA evaluation by eliminating the need for investigators to spend time sampling and evaluating posture from video camera recordings of workplace tasks. This study aims to determine whether it is possible to develop an automated system, based on a deep learning algorithm, that will reduce evaluation time and produce RULA score results that are sufficiently similar to those generated by manual evaluators in observational postural assessment.

2. Method

2.1. Data preparation

This study used Whole-Body Human Pose Estimation in the Wild, extending the MS COCO 2017 dataset that was manually annotated for 68 facial points, 42 hand points and 23 body and feet points, for training and validation of the proposed neural network (Jin et al., 2020). For each key point, the annotations indicate absolute horizontal and vertical distance from the top-left corner of the given image, as well as a numerical visibility flag assigned a value of 0, 1, or 2, where 0 is not visible and not available, 1 is available but not visible, and 2 is available and visible. The dataset can be found on the official repository, available at <https://github.com/jin-s13/COCO-WholeBody> (Jas and Fang, 2020). Based on RULA notation for whole-body postural analysis, this study used images with a visibility flag of 1 or 2 for the following 17 key points: nose (N), right eye (RI), left eye (LI), right shoulder (RS), left shoulder (LS), right elbow (RE), left elbow (LE), right wrist (RW), left wrist (LW), right trunk (RT), left trunk (LT), right knee (RK), left knee (LK), right ankle (RA), left ankle (LA), right knuckle (RN) and left knuckle (LN).

This study also used a pre-trained neural network to detect and crop a human figure from an original image to reduce the reception area for the postural detection network, thereby improving the accuracy of detection of the postural estimation network. The model weights and model architecture for the pre-trained neural network can be found at <https://github.com/experiencor/keras-yolo3>, available under open-source license. The class labels in the script were modified to detect only 'person' class. Data augmentation was applied to generate the vast amount of data required for training CNNs by rotating images at random angles and flipping each along its mid-vertical axis, as shown in Fig. 1. The images in the COCO dataset are of people in different postures engaged in common, real-life activities, which ensured that the model was trained on real data.

2.2. Network architecture

A regression based human pose estimation network was introduced to predict posture from images. The deep learning model in this study was drawn from a study by Toshev et al. (2013) and was trained using 17 key points from the recently published Whole-Body Human Pose Estimation in the Wild dataset (Jin et al., 2020). The input layer of the network accepts grayscale images of 128 x 128 pixels. Since the joint coordinates are in the absolute image coordinate system and the neural network input layer accepts images of 128 x 128 pixels, images were resized, and the label coordinates transformed relative to the new size of image. The images were normalized by subtracting the mean from each pixel across the channel and then dividing the result by the standard deviation, thereby speeding up convergence during network training.

$$X_s = \frac{X - \mu}{\sigma}$$

Equation (1) X is the pixel intensity, μ is the mean and σ is standard deviation of pixel intensity across entire channel.

Table 1 represents the architecture of the posture estimation network

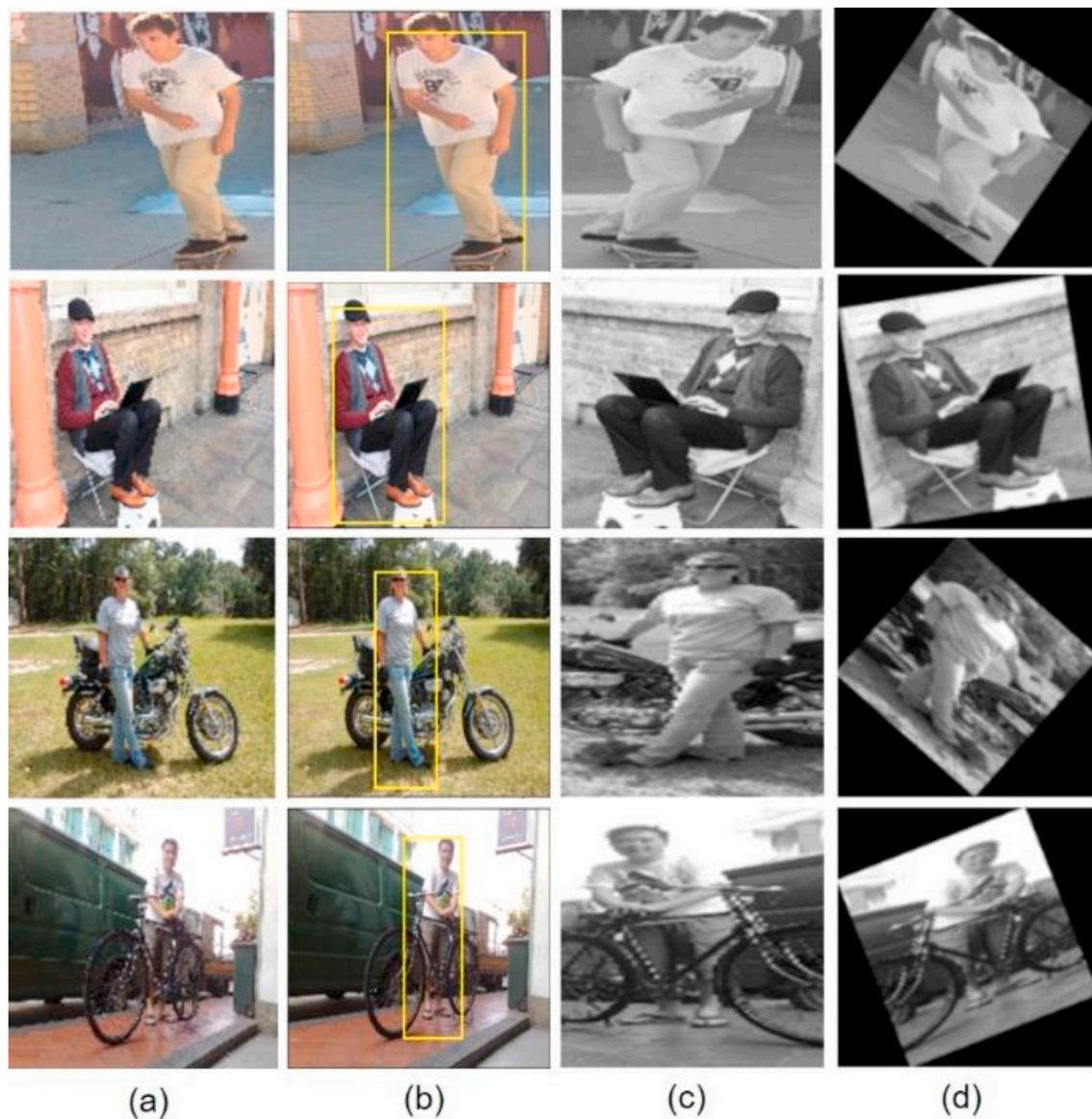


Fig. 1. (a) Sample training images from MS COCO dataset (b) Results of person detector model (c) Horizontal flip augmentation (d) Random rotation augmentation. Images used under CC by 2.0 license.

used in this study. The network consists of 5 convolutional layers for feature extraction where the first and second consecutive convolutional layers are stacked with a MaxPooling layer for reducing dimensionality of the feature map and decreasing the number of subsequent trainable parameters (Yamashita et al., 2018). Input to convolution layers were linearly transformed to have a zero mean and unit variance by employing batch normalization as part of the architecture (Ioffe and Szegedy, 2015). A Rectified Linear Unit was used as an activation function between each convolutional layer and mapped all negative inputs to zero (Maas et al., 2013). The output feature maps of the final convolutional layer were flattened and connected to 3 fully connected layers with 4096, 4096 and 1000 neurons, respectively, that were connected to the output layer of dimensions 34×1 . The dropout layer was implemented with a dropout regularization factor of 0.4 between each fully connected layer (Huang et al., 2017). The additional dropout layer reduced overfitting of the network and improved generalization by randomly dropping out neurons (Srivastava et al., 2014).

The performance of the model during forward propagation was calculated using Mean Squared Error (Sammut and Webb, 2011) as a

loss function, and learnable parameters were updated during back-propagation using Adam optimization (Diederik and Ba, 2015). The network was trained with a batch size of 64 and an initial learning rate of $1e-2$, which reduces by a factor of 2 up to $1e-8$ if the validation loss does not decrease for 6 consecutive epochs and which stops training when this condition is met to prevent the network from overfitting (Caruana et al., 2001). The capability of the network to generalize was assessed by the mean absolute error (MAE) of the model's performance on a validation set during training. MAE is the mean of the absolute difference between the predicted and absolute joint coordinates. To validate the model and to ensure that training and testing data are representative of the same sample, 20% of the dataset was reserved from the overall training set without any augmentation. The network was trained from scratch using a Google Cloud-based service called Google Colab with 25GiB GPU memory (Bisong, 2019).

2.3. RULA posture score estimation

RULA body posture scores were estimated from 2-D kinematic joint

Table 1

Architecture of postural estimation network.

Layer Type	Output Shape	#Parameters
Conv_2D_1(Convolution)	(None, 32, 32, 96)	11712
Bn_1(BatchNormalization)	(None, 32, 32, 96)	384
Max_Pooling_2D_1	(None, 16, 16, 96)	0
Conv_2D_2(Convolution)	(None, 16, 16, 256)	614656
Bn_2(BatchNormalization)	(None, 16, 16, 256)	1024
Max_Pooling_2D_2	(None, 8, 8, 256)	0
Conv_2D_3(Convolution)	(None, 8, 8, 384)	885120
Bn_3(BatchNormalization)	(None, 8, 8, 384)	1536
Conv_2D_4(Convolution)	(None, 8, 8, 384)	1327488
Bn_4(BatchNormalization)	(None, 8, 8, 384)	1536
Conv_2D_5(Convolution)	(None, 8, 8, 256)	884992
Bn_5(BatchNormalization)	(None, 8, 8, 256)	1024
Max_Pooling_2D_3	(None, 4, 4, 256)	0
Flatten	(None, 4096)	0
Dense_1	(None, 4096)	16781312
Bn_6(BatchNormalization)	(None, 4096)	16384
Dropout_1	(None, 4096)	0
Dense_2	(None, 4096)	16781312
Bn_7(BatchNormalization)	(None, 4096)	16384
Dropout_2	(None, 4096)	0
Dense_3	(None, 1000)	4097000
Bn_8(BatchNormalization)	(None, 1000)	4000
Dropout_3	(None, 1000)	0
Dense_4	(None, 34)	34034
Bn_9(BatchNormalization)	(None, 34)	136

locations obtained from the proposed deep learning model using Euclidean distance and the cosine of the angle between 2D vectors. Euclidean distance was used to calculate the distance between two joints or the length of a limb if the two joints belong to the same body part, and the inverse cosine was used to calculate angles between two limbs.

$$\theta_{ab} = \cos^{-1} \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

Equation (2) $\vec{a}$ and $\vec{b}$ are vectors with their respective heads pointing towards a joint and respective tails lying on the connecting limb.

Angles are denoted by θ with subscripts indicating the specific limbs on either side of θ as denoted by the first letter of the relevant limb, e.g., U for upper arm and L for lower arm, followed by a number to distinguish movement of single body part. For instance, flexion/extension movement is assigned the number 1, twist movement – 2, side-bending – 3, abduction/adduction movement – 5. S represents scores of a single body part, and its subscripts denote a specific limb and the movement of that limb in accordance with the numerical labels defined above. Fig. 2 and Fig. 3 represent RULA posture score calculations for each group from the 2D locations of body joints. This study has also incorporated thresholds for some parameters that are necessary to compute RULA scores as given by Vignais et al. (2017).

Upper Arm Flexion/Extension (θ_{U1}): Upper body joints (trunk and shoulder) and upper arm joints (shoulder and elbow) in sagittal plane were used to calculate upper arm flexion/extension angles.

Upper Arm Abduction (θ_{U5}): Upper Arm was considered elevated if the angle between upper body and upper arm was greater than 45° in the frontal plane.

Lower Arm Flexion/Extension (θ_{L1}): Upper arm joints and lower arm joints (wrist and elbow) in sagittal plane were used to determine lower arm flexion/extension angles.

Lower Arm Midline Posture (S_{L6}): Left and Right Wrist joints were tracked in frontal plane to estimate whether lower arms were working across the midline.

Wrist Flexion/Extension (θ_{W1}): Lower arm joints (elbow and wrist) and hand joints (wrist and knuckle) in sagittal plane were used to calculate wrist flexion/extension angles.

Upper Body Leaning (θ_{U4}): Upper Arm score was adjusted by –1 if

gravity assisted the posture and required to be initialized by investigator depending on the nature of the task.

Wrist midline posture (θ_{W6}): Lower arm joints (elbow and wrist) and hand joints (wrist and knuckle) in frontal plane were used to calculate wrist bending from midline posture. A score of +1 was added to the wrist score when this angle was inferior to –10° (radial deviation) or superior to 10° (ulnar deviation).

Neck Flexion/Extension (θ_{N1}): The neck key point is located at the center of left and right shoulder key points considering symmetry of human body. Upper body joints (trunk and shoulder) and head joints (neck and head) in sagittal plane were used to calculate neck flexion/extension angle.

Neck Twist (S_{N2}): The Euclidean distances between nose joints and left and right shoulder joints were used in the frontal plane to estimate neck side-twist.

Neck Side-Bending (θ_{N3}): We assumed that the head follows the same orientation as the line connecting the center of left and right eye to the mid-shoulder joints. Head joints (neck and head) were used to determine neck side-bending with respect to the vertical axis in the frontal plane.

Trunk Flexion/Extension (θ_{T1}): The mid-points of the right and left trunk key point were used to compute the mid-body trunk joint. The neck and mid-body trunk key point positions were used to calculate trunk flexion/extension with respect to vertical axis in the sagittal plane.

Trunk Twist (S_{T2}): Estimated by the Euclidean distance of shoulder joints from trunk joints in the frontal plane.

Trunk Side-Bending (θ_{T3}): Mid-upper body joint (trunk and neck) positions were used to calculate trunk side bend with respect to the vertical axis in the frontal plane.

Leg Posture (S_{L7}): Legs were assumed to be evenly balanced if the operator was in sitting posture or if the difference between right and left ankle y-coordinates was less than half of torso length in the frontal plane. Although Li and Xu (2019) used a threshold of 5 cm, using torso dimension is preferable because it considers the scale of the person in image.

Shoulder raise: As the worker did not raise his arm upward during the task, the ‘shoulder raising score’ was fixed at 0.

Force/Muscle use score: The force and muscle use scores need to be initialized in advance by the investigator according to the nature of task. For this study, muscle use score and load score were set to +1 because the posture was static, and the load intermittent.

2.4. RULA grand score estimation

The Group A score, as determined from upper arm, lower arm and wrist scores, and the Group B score, as determined from neck, trunk, and leg scores, combine with the muscle use score and force/load score to provide an overall posture score for both groups. Muscle use and load on the body parts were assumed to be constant throughout the work cycle. The scores from both groups were summarized into one action level, the grand score, that depicts the level of musculoskeletal risk associated with a given body posture and the corresponding priority of action associated with that risk, as indicated in Table 2.

2.5. Validation of the RULA posture score estimation

The posture angles obtained from the algorithm have been validated on 11 common occupational workplace posture images, as shown in Fig. 4, that have been manually evaluated in accordance with RULA by two ergonomics experts. One of the experts computed the angles between body parts for each posture manually by placing a digital goniometer directly on the subject's body during the task for each side, whereas another expert computed the angles from the test images using an online web angle measurement tool (Ginifab – Online Protractor). The images were taken from two different perspectives, one from the side to capture information about joint locations in the sagittal plane,

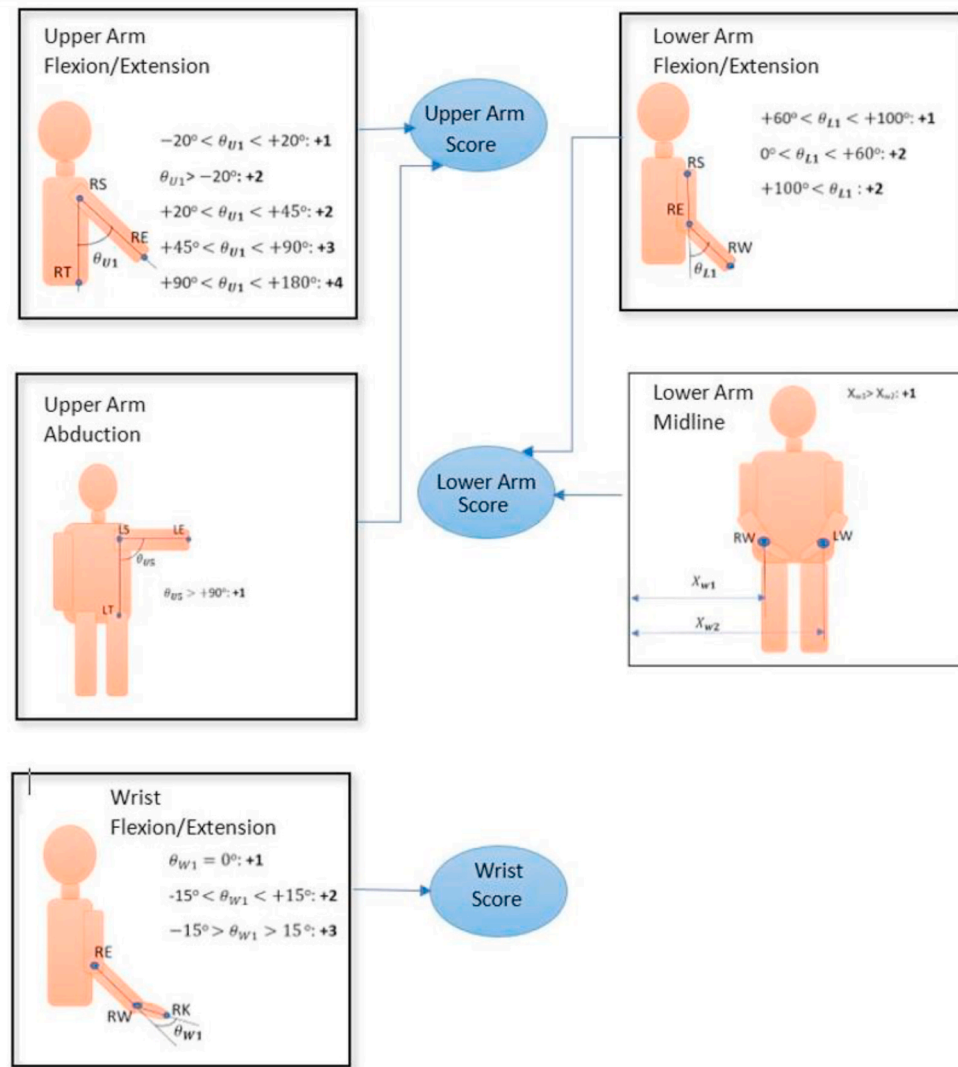


Fig. 2. Group A postural evaluation technique.

and the other from the front to capture information about joint locations in the frontal plane. Lowe et al. (2014) discusses the guidelines to record postures in a workplace for better quality and accuracy of analysis, and their guidelines have been followed in the current study for test postures in Fig. 4. The images were captured using a general-purpose camera (Panasonic Lumix DC-ZS70) and contain full body human postures in indoor and outdoor environments and performing different activities such as pushing, lifting, machining, and handling equipment.

The Intraclass Correlation Coefficient (ICC) and one-way ANOVA were applied to the collected data using SPSS software version 25.0 (IBM Corp., Armonk, NY). The ICC was calculated for the Group A score, Group B score and Grand Score for both left and right sides to assess the inter-reliability of the postural evaluation between the two rating methods, i.e., manual evaluation and the proposed algorithm. The ICC serves as a reliability index that reflects both degree of correlation and agreement between the results of two evaluation methods, comparing specifically the ratings from 2 ergonomics experts and those from the proposed machine learning algorithm in this study. The obtained ICC values were computed by a single-rating, absolute agreement, 2-way mixed effect model for postures shown in Fig. 4. This index is used when subjects are chosen at random, raters are fixed and difference among the ratings is considered relevant. A one-way ANOVA test was performed on RULA grand scores obtained by manual technique from the 2 experts (E1 and E2) and by the proposed algorithm to determine

whether any significant difference exists between the new method and the traditional manual approach.

3. Results

Fig. 5 plots the MAE loss for training and validation data computed at every epoch during training of the postural estimation network. The model trained for 187 epochs before stopping early to avoid overfit. The MAE loss computed at the end of final epoch on training data is 1.72, whereas the MAE loss on the validation data is 2.86. The trained model weights can be found <https://tinyurl.com/model-weights-fbp>.

Table 3 indicates the RULA evaluation for Group A, Group B and Grand Score along with the inference time of the algorithm for evaluating the postures. The table compares the evaluations from the two experts, denoted E1 and E2, and the results of the proposed automated technique, denoted A. Among the computed grand score results for both sides of the body posture, the proposed algorithm assigned the same evaluation scores those of the ergonomics experts to 40.91% of postures, whereas 36.36% of postures were assigned higher evaluation scores and the remaining 22.73% were assigned lower scores. The average inference time for evaluating the postures using the algorithm was 14.64 s.

As shown in Table 4, the ICC index for the Group A score is 0.776 for left-body posture and 0.867 for right-body posture, whereas the Group B score is 0.851 for both sides of the body. Similarly, the ICC index of the

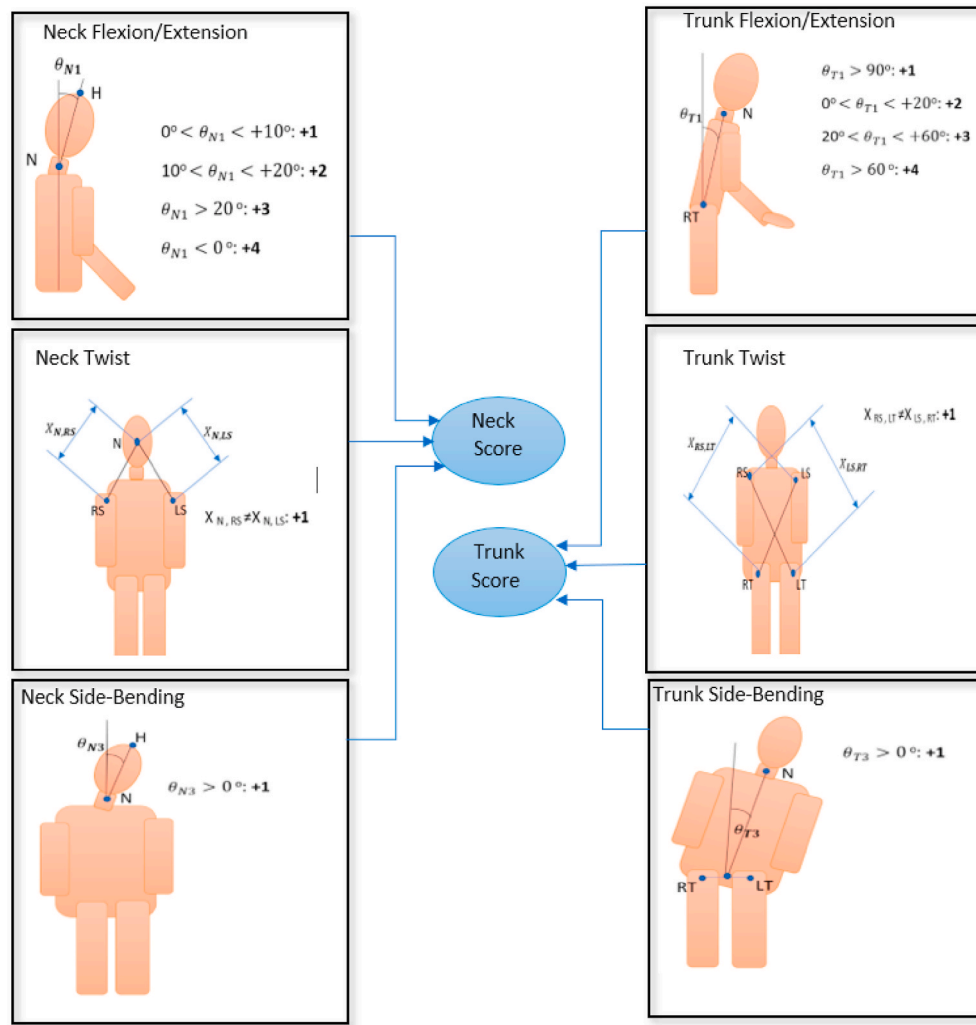


Fig. 3. Group B postural evaluation technique

Table 2
RULA grand score summary.

Grand Score	Level of WMSD Risk
1–2	Acceptable working posture if not maintained or repeated for long periods
3–4	Further investigation is needed, posture change may be required
5–6	Investigate and implement posture changes soon to avoid further exposure to WMSDs risk
7	Requires immediate attention and changes in posture

computed grand scores is 0.819 and 0.797 for left- and right-body posture, respectively.

As shown in Table 5, one-way ANOVA results revealed that the RULA grand scores from experts' respective postural evaluations were not significantly different ($p > 0.05$) from that of the algorithm with respect to left-body posture [F (Golabchi et al., 2015; Massaccesi et al., 2003) = 0.000, $p = 1.000$] and right-body posture [F (Golabchi et al., 2015; Massaccesi et al., 2003) = 0.392, $p = 0.679$].

4. Discussion

This study introduces a new method to estimate a RULA score from the 2-view (sagittal and frontal) body posture images with the help of a CNN-based neural model that is invariant in response to scale, visual

noise and color and can be used to assess postures in workplaces. The subject was recorded using two video cameras throughout the work cycle. One video-camera was positioned to capture information from the side, while another recorded the posture from the front. The performance of the proposed postural estimation network was evaluated by monitoring validation loss in non-augmented split training data. The results of the proposed method were compared with those of the traditional method to assess postural risks based on RULA assessment by computing the ICC as an index for agreement between the two techniques.

The model scored an MAE of 2.86 between predicted and absolute joint coordinates on the validation data, as represented in Fig. 5 which suggests little error in prediction. An overfitting is a condition in which the network learns the training data too well and subsequently becomes unable to generalize, which results in poor performance on new data. The proposed algorithm showed a difference in MAE of 1.14 between the training and validation data, which suggests very little likelihood of neural network overfitting. Table 3 displays the results of RULA evaluation of left- and right-side body posture using manual methods and the proposed algorithm. The proposed algorithm assigned the same evaluation scores as manual evaluators to 40.91% of postures. Although the proportion of identical evaluation scores is relatively low, the algorithm still assigned 77.27% of the postures the same or higher than corresponding scores achieved with the manual approach. Since postures with higher evaluation scores requires more attention from the evaluators, the results indicate that evaluation by the proposed method is more



Fig. 4. Example of test images of postures performing different kinds of occupational tasks from front and side perspective.

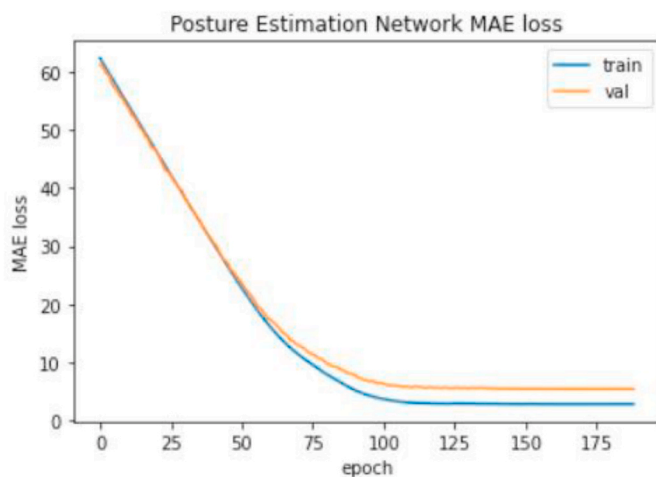


Fig. 5. MAE plot of posture estimation network during training.

conservative than that of the manual method and thus will have lower omission error than the manual method. This will lead to potentially risky postures being subjected to further investigation in order to minimize the risk of WMSDs in workplaces.

As shown in Table 4, the statistical analysis results demonstrate high levels of agreement between automated and manual evaluation techniques. The ICC values between 0.75 and 0.90 indicate good reliability, and values greater than 0.90 indicate excellent reliability (Koo and Li, 2016). The value of the ICC index is between 0.776 and 0.867 for both sides of body posture in this study, indicating that the algorithm's

estimated scores are in good agreement with the investigators' manual evaluation scores. For a few postures, the algorithm's RULA grand score was different from manual estimation for one side of the body posture. A possible explanation for this finding is that the upper body posture is asymmetrical between sides in the side-view image and/or that a given joint was occluded by other body parts. As mentioned in the literature on RULA, the selection of left- or right-side view can be made based on posture held for the greatest amount of the work cycle, location of the highest loads, or a subject's initially reported discomfort (McAtamney and Corlett, 1993). To offset uncertainty caused by asymmetrical posture between left- and right-side views, another camera can be mounted to record both side views.

A one-way ANOVA test was conducted on the RULA grand scores obtained from the two experts' manual evaluation and from the proposed algorithm's postural evaluation on both sides of the body. As per the results of the ANOVA test reported in Table 5, there was no significant effect of the evaluation technique on the final RULA Grand score assessment results ($p > 0.05$). This suggests that the proposed technique can be used as an alternative to the traditional, manual approach to evaluate posture based on RULA.

The time taken to evaluate a posture manually is based on factors like posture complexity and investigator experience, whereas the machine learning technique is independent of such factors. As depicted in Table 3, the range of inference time for the postural estimation algorithm is 13.68–16.45 s. The proposed algorithm thus drastically reduces the time taken for postural evaluation and helps ergonomic investigators to make quick decisions. The reported inference time is dependent on the size of input image and GPU of the machine in which the model is running, which explains the slight difference.

This study makes several contributions to workplace safety and the literature that studies it. The present study is the first to implement a

Table 3

Group A, Group B and Grand Score from RULA postural evaluation for left and right sides of body posture by Expert 1 (E1), Expert 2 (E2) and postural estimation algorithm (A).

Posture	Group A Score						Group B Score						Grand Score						Time (in seconds)	
	Left			Right			Left			Right			Left			Right				
	E1	E2	A	E1	E2	A	E1	E2	A	E1	E2	A	E1	E2	A	E1	E2	A	A	
1	2	2	2	2	2	2	3	3	2	3	3	2	5	5	4	5	5	4	14.01	
2	3	3	2	3	3	3	2	2	5	2	2	5	5	5	6	5	5	7	13.83	
3	3	3	2	4	4	3	5	5	5	5	5	5	7	7	6	7	7	7	13.68	
4	4	4	3	4	4	4	3	3	3	3	3	3	6	6	6	6	6	6	13.77	
5	3	3	2	3	3	3	6	6	6	6	6	6	7	7	6	7	7	6	14.76	
6	2	2	2	2	2	2	2	2	2	2	2	2	4	4	4	4	4	4	14.22	
7	3	3	3	3	3	3	3	3	3	3	3	3	5	5	6	5	5	6	14.26	
8	3	3	2	3	3	2	3	3	4	3	3	4	5	5	6	5	5	6	14.29	
9	4	4	4	4	4	5	2	2	3	2	2	3	6	6	6	6	6	7	16.45	
10	4	4	4	2	2	2	2	2	2	2	2	2	6	6	6	4	4	4	16.01	
11	4	4	3	2	2	3	5	5	6	5	5	6	7	7	7	6	6	7	15.76	

Table 4

ICC index for Group A Score, Group B Score and Grand Score with a confidence interval of 95% for left and right-side body postures.

Measure	Intraclass Correlation Coefficient (ICC)	
	Left-body Posture	Right-body posture
RULA Group A Score	0.776	0.867
RULA Group B Score	0.851	0.851
RULA Grand Score	0.819	0.797

Table 5

Results from independent sample *t*-test (Traditional Vs. Proposed Method Evaluation).

Variables	Manual Evaluation		Proposed Method Evaluation		<i>t</i>
	Mean	SD	Mean	SD	
RULA Grand Score - Left	5.73	1.009	5.73	0.905	0.00
RULA Grand Score - Right	5.45	1.036	5.82	1.250	-0.743

machine learning algorithm for observation-based RULA posture assessment to estimate all relevant body posture scores, including the wrist score, from images of postures in real workplace (Peppoloni et al., 2015; Plantard et al., 2017; Clark et al., 2012; Li et al., 2020). Additionally, the output of the algorithm can be directly interpreted to assess the risks associated with postures, reducing the time and cost typically required to train an ergonomic analyst in conducting a RULA assessment survey. Further, with this method, investigators will no longer need to segment body parts or evaluate posture manually, which will reduce evaluation time and eliminate human error. In addition, the proposed system is easy to use, allowing an investigator to assess postures quickly at the workplace without conducting preliminary surveys to only select postures that were reported for discomfort. Based on our results, we conclude that the proposed system will aid ergonomics experts in carrying out the investigation at the workplace and would encourage the use of advanced tools for ergonomic assessments.

There are several features and limitations to this study that should be noted. The first limitation of this study is that the reliability of the algorithm is dependent on the recordings of the postures. Video images represent posture 2-dimensionally and may cause perspective errors if the camera view is not perpendicular to motion of interest thus, the postures must be captured such that images depict true location of body joints (Lowe et al., 2014). The proposed computer vision model can estimate individual group scores from a single view image but requires 2 images in perpendicular plane to avoid parallax error in estimating true joint positions and to compute RULA grand scores for postures. Two ergonomic analysts can record postures for the tasks that require

movement more than translation in one direction or a 1 view image can be used to obtain corresponding joint movement such as upper arm flexion/extension from side view image or neck side-bending/twist movement from front view image. The image view required for computing the joint movement is explained in detail in section 2.3. In the future, more advanced neural networks can be incorporated to predict 3D joint locations from the raw image pixels, thereby eliminating the need for images taken from two different orientations (Martinez et al., 2017; Zhou et al., 2017; Li and Chan, 2015). The second limitation of the study is that the proposed machine learning algorithm is dependent on the person detector model to perform body joint predictions. The person detection model fails on images in which body postures are obstructed heavily with objects. Fig. 6 shows examples of such images. To overcome this limitation, the posture must be recorded from a better position, which will reduce occlusion and aid the person detector model in detecting the subject. Although it is recommended to have camera view perpendicular to the plane of interest for accurate assessment of the angle, the investigator can still alter the position of recording cameras to his own judgement in case the workplace occludes a subject's posture. For instance, the posture in Fig. 6 is occluded from the vehicle frame to such an extent that the person detector model is unable to detect the subject. However, the same posture can be captured from a different position to obtain results, as shown in Posture 5 in Fig. 4.

The results of the proposed method were validated against ratings by ergonomic analysts who evaluated the postures using basic ergonomics



Fig. 6. Example of posture that the person detection model failed to detect due to high object obstruction.

tools (goniometer and online angle measurement software) to simulate the practical study in the workplace. Although this validation approach could be questioned, this choice was made because recent surveys have proved that ergonomic practitioners worldwide are still prefer to use basic tools over direct measurement tools to evaluate postures in workplaces (Lowe et al., 2019).

The observation-based posture evaluation method, despite being popular among ergonomics practitioners, is prone to omission errors that can be effectively overcome with the use of the method deployed in this study. The proposed algorithm has proven to be more conservative in posture evaluations than manual evaluations, which will lead investigations for ergonomic intervention to discover more postures that pose WMSD risks. The proposed method can evaluate postures without omission error and with performance similar to but faster than manual evaluations. This method, therefore, can aid ergonomic evaluators in conducting RULA-based surveys of occupational postures in workplace conditions.

The posture estimation network discussed in this study can be easily extended to other posture assessment tools like ALLA, REBA and OWAS that evaluate body postures to determine their associated WMSD risks. For instance, REBA evaluates body postures in a similar way to the proposed method by estimating the angles of leg, trunk, neck, arm and wrist. The output of the postural estimation network is a 2D body joint location that can be processed with a python-based script to evaluate postures based on RULA. This separation of logic allows the proposed postural estimation network to be used for evaluation of postures based on other tools.

5. Conclusion

This study discusses a deep learning technique to estimate posture scores from images of postures at occupational workplaces by detecting 2D coordinates of body joints, including wrist joints, to evaluate work postures based on RULA. This study proposes a posture estimation network to detect body key point locations of work postures from images. The accuracy of the network has been validated on a subset of training images and achieves high precision. The network has been tested on common workplace postures recorded in indoor and outdoor environments and on workers performing different activities such as lifting, lowering, pushing, etc. The input images are taken from two perspectives, the front and side of a subject's posture, depicting the true posture in 3D space. The results of the algorithm were compared with those from manual evaluation by 2 ergonomics experts and indicated very good agreement on an ICC index. The automation of observation-based techniques for posture assessment using deep learning will produce results with similar performance as manual evaluators, eliminate errors due to human mistakes and reduce time for posture evaluation in workplaces. In the future, advanced machine learning algorithms will be developed to estimate postures in 3D space from a single image. To further this end, the present study can be extended to evaluate postures based on different postural screening tools.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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